

Kumar Manglam Pandey

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Work authorization: Canada (Study Permit, PGWP-eligible) | Requires sponsorship for U.S. roles

SUMMARY

ML engineer with 8+ years production software experience (founding engineer, technical consulting) transitioning into applied AI/ML research and development. Currently building agent evaluation infrastructure as a Machine Learning Associate at Vector Institute x COHUMAN Labs.

PROFESSIONAL EXPERIENCE

Machine Learning Associate — Vector Institute x COHUMAN Labs

Aug 2026 - Present

- Prototyping a two-step AI agent review system evaluating code and configuration artifacts against financial regulations, with drift monitoring and audit-trail logging for regulatory traceability by applying LLM fine-tuning and agentic architectures targeting on secure, on-premise infrastructure.

HAIVE Graduate Intern (Machine Learning & Risk) — RISC Foundation

Aug 2026 - Present

- Assessing algorithmic fairness in Ontario Auto Insurance pricing model — spearheading a comprehensive literature review to identify data requirements and statistical metrics. Reviewing Ontario-specific regulatory frameworks while collaborating in a hybrid, interdisciplinary Actuarial and Data Science team alongside industry risk executives and academic mentors.

Project Manager — Partnered with Youth Service Bureau and HICC

Jun 2026 | C4 - Cross Campus Capstone Classroom

- Led a 5-person cross-functional team to design “Third Space”, an experiential-learning program for youth transitioning out of transitional housing, replacing online coursework with real-world scenario simulations, designed and proposed in one month in partnership with Youth Service Bureau and HICC.

Co-Founder — TISC Research & Development Pvt. Ltd.

Oct 2020 – Dec 2022

- Co-founded and led a 12-person team building SolaReady, an end-to-end solar-adoption platform; scaled to 100+ paying clients through India's PM Rooftop Solar Programme.

Technical Consulting Engineer — Cisco Systems

Dec 2018 – Oct 2020

- Diagnosed and resolved complex routing, performance, and OS-level issues across large-scale enterprise, service-provider, and government networks; recreated customer environments in lab/cloud setups to validate fixes, and maintained SLA compliance.
- Developed Python automation scripts to parse large log datasets, monitor key risk indicators across hundreds of devices, correlate alerts, and trigger diagnostics on anomalous CPU/memory conditions; reduced manual analysis effort and average resolution time by ~30%.

Founding Engineer — CubeK Technologies Pte. Ltd.

Dec 2016 – Dec 2018

- Founding engineer for electronicscrazy.sg; built the full B2B/B2C e-commerce platform from scratch, scaling to tens of thousands of users and ~80-100 daily orders within the first month; integrated payments, logistics, and analytics APIs.

PROJECTS

Doctor Scheduling Agent - Agentic AI System

Jun 2026 | github.com/manglam8/doc-schedule-agent | doc-schedule-agent.streamlit.app

- Built a production-deployed agents workflow enabling patients to book clinic appointments through a natural-language chat interface, using a LangGraph ReAct loop (LLM → tool dispatch → LLM) with three LLM-callable tools: speciality search, availability check, and appointment booking.
- Architected a full-stack system with FastAPI backend, SQLite (WAL mode) persistence, SQLAlchemy ORM, and Streamlit frontend; implemented two-layer double-booking prevention via DB-level UNIQUE constraint and agent-side IntegrityError recovery with automatic slot regeneration.

- Deployed on Streamlit Cloud with 17 unit tests (tool-level and graph routing) using an in-memory fixture database; integrated Gemini 2.5 Flash Lite via LangChain with InMemoryRateLimiter enforcing API quota compliance.

FairHire - Bias Auditing Tool for Hiring Data

Mar 2026 | github.com/manqlam8/fairhire | fairhire.onrender.com

- Built an end-to-end bias auditing pipeline for structured hiring datasets using FastAPI, PostgreSQL, and SQLAlchemy, deployed on Render with a persistent audit trail tracking risk indicators, fairness metrics, and debiasing interventions across iterations.
- Implemented a fairness engine computing EOD, Demographic Parity Difference, and Equalized Odds across demographic groups, with severity thresholds that classify findings by risk level and automated debiasing recommendations including SMOTE resampling and threshold optimization.

Search-and-Rescue Optimization via Cost-Oriented POMDP

Mar 2026 | *Advanced Computer Vision — York University*

- Formulated a decision-making algorithm for autonomous search-and-rescue scenarios using a Partially Observable Markov Decision Process (POMDP) framework.
- Optimized epistemic action selection between Active Vision and Manipulation strategies based on cost modeling, enabling an agent to minimize uncertainty while operating under resource constraints.

Fairness Analysis & Debiasing on CCHS Dataset for Mental Health Screening

Mar 2026 | *Fairness & Bias in AI — York University* | [Fairness Bias Tutorial](#)

- Analyzed fairness metrics (EOD, DPD, EqOdd) on the Canadian Community Health Survey 2022 dataset using decision tree and neural network models to quantify discriminatory bias across: demographic groups, age groups, indigenous & non-indigenous group, immigrant/non-immigrant, disability, and health regions.
- Implemented and compared two debiasing methodologies: SMOTE Resampling and Threshold Optimization, evaluating trade-offs between model performance (Accuracy, F1, AUC, Precision, and Recall) and fairness outcomes.

RESEARCH EXPERIENCE

Research Assistant — EXINES Lab, York University

May 2026 – Aug 2026 | Supervisor: Professor Maleknaz Nayeji

- Conducted research on SWE-bench Multimodal evaluation of AI coding agents on complex software engineering tasks like bug fixing and troubleshooting.

Research Assistant — ARIN Lab, York University

Apr 2026 – Aug 2026 | Supervisor: Professor Ines Arous

- Supported RLHF research on LLM alignment by optimizing human feedback; ran training experiments on H100/SLURM infrastructure, and implemented reward-model pipelines with real time visualization on Weights & Biases.

EDUCATION

York University — Master of Science, Computer Science (AI Specialization)

Jan 2026 – Present | Toronto, ON | Anticipated Graduation: Dec 2026

- Relevant Coursework: Advanced Computer Vision, Fairness & Bias in AI, Machine Learning Theory, Human Perception and Performance in HCI, Data Mining, Robotics
- Academic Standing: Maintained straight 'A' average across core graduate coursework.

SKILLS

Programming & Data: Python, PyTorch, Pandas, NumPy, Scikit-learn, TensorFlow, PostgreSQL, SQL, Bash

Machine Learning & AI: Supervised & Unsupervised Learning, Neural Networks, Computer Vision, NLP, LLM, Transformers, HuggingFace, Prompt Engineering

Agentic AI & LLM Frameworks: LangGraph, LangChain, FastAPI, Streamlit, OpenAI/Gemini APIs, ReAct Agents

Tools & Infrastructure: H100 GPU, SLURM, Digital Research Alliance of Canada's Fir & Narval Supercomputers, Jupyter Notebooks, Google Colab, Git, Linux, Docker, REST APIs, GCP

Project & People: Problem Solving, Partner/Client Consultation, Research, Planning and Execution, Leadership, Teamwork and Collaboration, Communication, Conflict Resolution